Feature ID: F-6.6.2

Protective Safety Mode (PSM)

User Problem Solved

Platform infrastructure must remain operational and trustworthy even under extreme stress: regional surges, system overload, connector failures, or security incidents. Users should never experience silent data loss, unpredictable behavior, or corruption during platform degradation.

Protective Safety Mode ensures the platform **degrades gracefully** rather than failing catastrophically. When activated, PSM restricts platform capabilities to **core, proven-stable operations** while blocking risky mutations that could compromise data integrity.

In one sentence: "When something goes wrong with the platform, we automatically switch to a safer mode where critical functions still work and nothing breaks silently."

Purpose

Protective Safety Mode is a **platform-wide operational state** (not a user-facing feature flag) that activates automatically when the platform detects conditions threatening data integrity, performance, or trustworthiness.

PSM serves three goals:

1. **Prevent Silent Failures**: Block operations that might corrupt data, fail invisibly, or produce unreliable outputs during degraded conditions
2. **Preserve Core Capabilities**: Keep essential user workflows functional (trip creation, incident capture, evidence upload, **evidence export**) even when advanced features unavailable
3. **Enable Recovery**: Surface degraded state clearly to users and operators; emit recovery telemetry to restore normal operation

PSM is **not** a user-requested mode. Users do not "turn on" PSM. The platform enters PSM automatically based on governance triggers (Section 3.0).

Regional Scoping: PSM activates **per region**, not platform-wide. US-East can be in PSM while EU-West operates normally.

In-Scope Behaviors

1. Automatic PSM Activation (Governance Triggers)

PSM activates automatically **per region** when **any** of the following conditions are detected:

Trigger	Condition	Expected Recovery Time	Escalation Threshold	Source
Event Logging Unavailable	Append-only event ledger cannot write events (3 consecutive failures)	10 seconds (auto-retry)	60 seconds → P1 alert	Section 3.0, 8.4
Regional Overload	Incident volume >10,000/hour in single region (5x normal baseline)	30-120 minutes (traffic subsides)	4 hours → P2 alert	Section 15.4
External Data Sources Blocking Us	3+ critical connectors down simultaneously (e.g., American Airlines, United, Amadeus all blocking)	24-48 hours (airline websites stabilize)	72 hours → P3 alert	Section 15.5
Database Backup Sync Stopped	Database replication failure >60 seconds (primary-to-secondary sync failed)	10-30 seconds (auto-failover)	60 seconds → P1 alert	Section 15.6
AI Quality Below Safety Threshold	AI output confidence distribution falls below safety threshold (>30% low-confidence responses)	5-15 minutes (AI tier escalation or model restart)	30 minutes → P2 alert	Section 9.2
Unauthorized Platform Changes Detected	Mutation guard bypass attempt detected (someone trying to change data without proper authorization)	Immediate block (no auto-recovery)	Immediate → P1 alert	Section 3.0

Cost Explosion Risk	Cost projection exceeds safety threshold without authorization (AI spend spiking)	Manual review required	Immediate → P2 alert	Section 10.2

Activation Logic:

sql
```sql
FUNCTION enter_protective_mode(
  region_id TEXT,
  trigger_reason TEXT
) RETURNS VOID AS $$
BEGIN
  -- Set regional mode
  UPDATE platform_state
  SET operational_mode = 'PROTECTIVE',
      protective_mode_reason = trigger_reason,
      protective_mode_entered_at = now()
  WHERE region = region_id;

  -- Emit governance event
  INSERT INTO event_ledger (event_type, severity, region, payload)
  VALUES ('protective_mode_entered', 'critical', region_id,
          jsonb_build_object(
            'reason', trigger_reason,
            'timestamp', now(),
            'region', region_id
          ));

  -- Disable risky features for this region
  UPDATE feature_flags
  SET enabled = FALSE
  WHERE feature_id IN ('deep_scan', 'paid_unlock', 'claim_packet_generation')
    AND region = region_id;

  -- Notify operators (PagerDuty/Opsgenie)
  PERFORM notify_operator_alert(
    'SAFE MODE ACTIVE',
    format('Region %s entered Safe Mode: %s', region_id, trigger_reason),
    region_id
  );
END;
$$ LANGUAGE plpgsql;
```

Entry Event:

- *protective_mode_entered* (severity: critical, includes region_id and trigger reason)

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2. Restricted Operations (What's Blocked in PSM)

When a region is in PSM, the platform **blocks** the following high-risk operations **for users in that region**:

Blocked Mutation Operations:

- ❌ Deep Scan creation (Section 3.2) — AI-intensive, expensive, unreliable under degraded conditions
- ❌ Paid unlock (Section 10.2) — Payment processing risky; revenue integrity uncertain
- ❌ Claim packet generation — AI-heavy, must be high-confidence
- ❌ Identity mutations (email change, password reset via automated flow) — Security-sensitive
- ❌ Evidence deletion — Data integrity risk
- ❌ Policy attachment (if connector down) — Coverage graph unreliable
- ❌ Group invite acceptance (new participants) — Permissioning complexity

Blocked Analytical Operations:

- ❌ Coverage Graph Snapshot creation (if connectors unavailable) — Unreliable data
- ❌ Financial modeling (Section 10.5) — Cost projections untrustworthy under surge
- ❌ AI-powered activity suggestions (F-6.5.19) — AI tier escalation unavailable

Rationale: These operations are either:

1. AI-intensive (rely on external LLM tiers that may be unavailable)
2. Financially sensitive (revenue/cost integrity uncertain under degraded state)
3. Data-mutation-heavy (risk of corruption if ledger/database unreliable)

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3. Allowed Operations (What Still Works in PSM)

Core User Workflows (never blocked, even in PSM):

✅ **Trip creation** (Section 3.5) — Core capability, never paywalled ✅ **Trip draft editing** (F-6.5.17) — Users can still plan travel ✅ **Itinerary confirmation** — Users can finalize trips ✅ **Incident capture** (voice or manual) — Critical safety workflow ✅ **Evidence upload** — Preserve user data even if processing deferred ✅ **Evidence export** (Section 8.7) — Users can download

their documentation anytime ✅ **Voice narration** (Section 7.4) — Transcript saved (immutable), parse deferred ✅ **Emergency Contact access** (F-6.6.5) — Safety-critical, always available ✅ **Account login/logout** — Identity access maintained ✅ **Read-only access** (view trips, view evidence, view policies)

Degraded Operations (work with warnings):

Route normalization (F-6.5.18) — Runs in degraded mode (no external API calls) **Activity suggestions** (manual only) — AI-powered suggestions disabled; user can still add manually **Onboarding** — Allowed, but paid plan upgrades deferred

Rationale: Core intake workflows (trip creation, incident capture, evidence preservation, **evidence export**) must always function. PSM allows these while blocking operations that depend on unreliable external systems or AI processing.

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4. User-Facing Communication (Calm, Transparent Degradation)

When a user attempts a **blocked operation** while PSM is active, they see:

Copy (Deep Scan Blocked):

Header: "We're running in safe mode right now"
Detail: "Deep Scan is temporarily unavailable while we stabilize the platform. Your trip data is safe, and you can still add details manually."
CTA: "Got it"
Secondary: "Learn more about safe mode"

Copy (Paid Unlock Blocked):

Header: "Upgrades are paused for now"
Detail: "We've temporarily paused paid upgrades to ensure reliable service. You can still create and edit trips for free."
CTA: "Continue with free tier"
Secondary: "Notify me when available"

Copy (Claim Packet Generation Blocked):

Header: "Claim packets aren't available right now"
Detail: "We're running in safe mode, so claim packet generation is temporarily unavailable. You can still upload evidence and export your documentation."
CTA: "Upload evidence"
Secondary: "Export my documentation"

General PSM Banner (shown at top of all screens while PSM active):

```
Banner: "    Safe mode active — Some features temporarily unavailable"
Tap to expand: "The platform is running in safe mode to protect your data during high load.
Core features (trip planning, incident capture, evidence upload, evidence export) are fully
functional. Advanced features will return when the platform stabilizes."
```

Key Principles:

- ✅ Never use the word "error" or "broken"
- ✅ Frame as **protective** (we're keeping your data safe) not punitive (you can't do this)
- ✅ Always offer alternative path (manual workflows, export still available)
- ✅ Never apologize for PSM activation (it's working as designed)

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5. Operator Interface (FOCL Controls) — Founder-Friendly Language

FOCL Panel: Regional Safe Mode Status

Operators (Founder only) view regional status and manage resources (not "override PSM"):

FOCL Screen (Founder-Friendly):

```
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Platform Health Overview
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Region Status:

  US-East: SAFE MODE ACTIVE (2 hours, 15 minutes)
  Issue: Regional overload — 12,000 incidents/hour (5x normal)
  Active Features: Trip creation, evidence upload, incident capture
  Blocked Features: Deep Scan, claim packets, paid unlocks
  Auto-Recovery: Expected in 30-90 minutes (traffic subsiding)

  EU-West: NORMAL
  All features operational

  Asia-Pacific: NORMAL
  All features operational

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```

```
US-East Resource Allocation:

  Current Capacity:
    • API requests/hour: 50,000 (at 96% utilization)
    • AI Tier 1 (internal): OFFLINE (GPU Server #2 crashed, restarting)
    • AI Tier 2 (Gemini Flash): ACTIVE (fallback mode, costs +$12/hour)
    • Database: Primary operational, backup sync recovered

  Resource Options:
    [ ] Allocate +50% API capacity (2 hours, auto-revert) — Cost: +$45/hour
    [ ] Enable emergency Deep Scan queue (100 requests max, Tier 3 AI only)
    [ ] Manually approve paid unlocks (one-by-one review required)

  Feature Toggles (Time-Boxed):
    [ ] Enable Deep Scan (1 hour, Tier 3 AI only) — Risk: cost spike
    [ ] Enable claim packet generation (30 minutes, manual queue) — Risk: quality degradation
```

```
Incident Timeline (US-East):

  14:30 UTC: Regional surge detected (10,500 incidents/hour)
  14:31 UTC: Safe Mode activated automatically
  14:32 UTC: Deep Scan disabled, paid unlocks disabled
  14:45 UTC: GPU Server #2 (US-East-1 cluster) crashed
  14:46 UTC: AI Tier 1 offline, failover to Tier 2 (Gemini Flash)
  15:00 UTC: Traffic peak: 12,300 incidents/hour
  16:15 UTC: Traffic declining: 9,200 incidents/hour
  16:30 UTC: GPU Server #2 restarted, health check passing

  Auto-Recovery Projection: 17:00-17:30 UTC (traffic < 8,000/hour + GPU stable)
```

Founder-Friendly Trigger Narratives:

Trigger (Technical)	Founder-Friendly Narrative
ledger_write_failure	"Event logging unavailable — protecting data integrity"

`regional_surge`	"Regional overload: 12,000 incidents/hour (5x normal traffic)"
`connector_instability_cascade`	"American Airlines, United, Amadeus all blocking us — manual entry needed"
`audit_log_saturation`	"Audit log system overloaded — pausing non-critical operations"
`ai_confidence_skew`	"AI quality below safety threshold — using more expensive backup systems"
`mutation_guard_bypass`	"Unauthorized platform changes detected — all mutations blocked"
`cost_projection_exceeded`	"AI spending spike detected — pausing expensive operations"

Resource Allocation Controls (NOT "Override PSM"):

- **Allocate capacity**: Add temporary API/GPU resources to specific region
- **Enable emergency queues**: Limited Deep Scan requests (Tier 3 AI only, cost-aware)
- **Time-boxed feature toggles**: Enable specific features for limited duration (auto-revert)

NO "Exit Safe Mode" Button — System auto-recovers when conditions clear.

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6. PSM Exit Conditions (Automatic Recovery)

PSM exits automatically **per region** when **all** of the following conditions are met:

1. **Trigger condition resolved** (e.g., ledger writes successful, surge subsided, connectors restored)
2. **Recovery validation passed** (platform runs self-check: can write to ledger, AI tiers responsive, connectors operational)
3. **Cooldown period elapsed** (5 minutes minimum after trigger resolution to prevent flapping)

Exit Logic:

sql

```sql
FUNCTION evaluate_protective_mode_exit(region_id TEXT) RETURNS VOID AS $$
DECLARE
  v_trigger_resolved BOOLEAN;
  v_recovery_validated BOOLEAN;
  v_cooldown_elapsed BOOLEAN;
BEGIN
  -- Check if trigger condition resolved
  v_trigger_resolved := check_trigger_conditions_clear(region_id);

  -- Run recovery validation
  v_recovery_validated := run_recovery_checks(region_id);

  -- Check cooldown (5 minutes since trigger last detected)
  SELECT (now() - MAX(detected_at)) > INTERVAL '5 minutes'
  INTO v_cooldown_elapsed
  FROM protective_mode_triggers
  WHERE region = region_id AND resolved_at IS NULL;

  -- Exit PSM if all conditions met
  IF v_trigger_resolved AND v_recovery_validated AND v_cooldown_elapsed THEN
    UPDATE platform_state
    SET operational_mode = 'NORMAL',
        protective_mode_exited_at = now()
    WHERE region = region_id;

    -- Re-enable features
    UPDATE feature_flags
    SET enabled = TRUE
    WHERE feature_id IN ('deep_scan', 'paid_unlock', 'claim_packet_generation')
      AND region = region_id;

    -- Emit recovery event
    INSERT INTO event_ledger (event_type, severity, region, payload)
    VALUES ('protective_mode_exited', 'info', region_id,
            jsonb_build_object(
              'duration_minutes',
              EXTRACT(EPOCH FROM (now() - (
                SELECT protective_mode_entered_at
                FROM platform_state
                WHERE region = region_id
              )))/60,
              'region', region_id
            ));
```

```
  END IF;
END;
$$ LANGUAGE plpgsql;
```

Exit Event:

- *protective_mode_exited* (severity: info, payload includes duration and region)

User-Facing:

- PSM banner disappears
- Toast notification: "Platform fully restored — All features available"

Founder-Facing (FOCL):

```
Region US-East: SAFE MODE EXITED
Duration: 2 hours, 47 minutes
Trigger: Regional overload (12,300 incidents/hour peak)
Resolution: Traffic subsided to 7,800/hour, GPU Server #2 stable
All features re-enabled at 17:22 UTC
```

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7. If Founder Unavailable (48-Hour Survivability + Emergency Delegation)

Platform Survivability Doctrine (Sections 15.0, F-6.5.16, 7.3):

- Platform must survive 24-48 hours without Founder login
- Core workflows remain functional
- Auto-recovery mechanisms operate autonomously
- Manual intervention only required for non-critical decisions

Emergency Mode Escalation (Section 8.4.8):

If PSM escalates to **Emergency Mode** (≥2 regions in PSM simultaneously, or ledger integrity failure):

- **Duration**: Max 4 hours
- **Founder notification**: Push + SMS + Email (all sent simultaneously at T+0)
- **24-hour review requirement**: Founder must review within 24 hours
- **T+30min**: If Founder unreachable, Standing Containment Extension activates:
    - PSM maintained
    - New export requests queued (not executed)
    - New user registrations held in pending state
    - Legal counsel auto-notified: "Founder unreachable — Standing Containment Extension active"

After 48 Hours (if conditions haven't cleared):

- **Auto-recovery attempts continue** (Kubernetes restarts, AI tier failover, connector retry)
- **System stays in Safe Mode indefinitely** (safe default)
- **Emergency Delegate notified** (F-6.6.6): Backup contact receives read-only status update
- **No auto-exit from PSM** — System waits for conditions to clear OR Founder manual approval

Implied Behavior (Doctrine):

- System never auto-escalates permissions
- System never auto-exits PSM if conditions haven't cleared
- Legal counsel remains notified (T+30min notification stands)
- Platform continues functioning in degraded state (core workflows operational)

Out-of-Scope Behaviors

1. User-Controlled PSM

❌ Users cannot manually activate or deactivate PSM.

Rationale: PSM is a governance-level safety mechanism, not a user preference.

2. Platform-Wide PSM (Non-Regional)

❌ PSM is **per-region** at MVP. No global PSM toggle.

Phase 2: Cross-region orchestration (if US-East and EU-West both in PSM → escalate to Emergency Mode).

3. PSM for Non-Platform Conditions

❌ PSM does NOT activate for:

- Single user experiencing app crash
- Localized network connectivity issues
- Individual connector timeout (unless part of cascade)

Rationale: PSM is for **regional or platform-wide** degradation, not user-specific issues.

Inputs

Input	Source	Format	Required
Trigger condition	Platform monitoring	Event signal	YES
Region ID	Load balancer / request metadata	String (e.g., "US-East")	YES
Resource allocation request	FOCL (Founder)	Manual action	NO
Recovery validation results	Self-check RPC	Boolean	YES (for exit)

Outputs

Output	Format	Destination	Conditions
protective_mode_entered	Event	event_ledger	PSM activation (includes region)
protective_mode_exited	Event	event_ledger	PSM exit (includes region, duration)
protective_mode_resource_allocated	Event	event_ledger	Founder allocated resources
User-facing banner	UI component	All screens (region-scoped)	While PSM active
Blocked operation message	Modal/toast	User screen	Attempted blocked action

FOCL status panel	UI	FOCL dashboard	Always visible (all regions)
Founder notification	Push/SMS/Email	Founder contact registry	Emergency Mode (≥2 regions in PSM)
Emergency delegate notification	Email	Delegate contact	T+48hr if Founder unreachable

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Dependencies

Sections

- **Section 3.0** (Governance Substrate) — PSM trigger matrix
- **Section 3.5** (Trip Maturity States) — PSM mutation guards
- **Section 8.4** (Permissioning & Audit Logs) — Resource allocation logging
- **Section 8.7** (Export Governance) — Export allowed in PSM
- **Section 8.8.10** (Founder Reach Protocol) — Emergency notification sequence
- **Section 9.2** (Confidence Labels) — AI safety thresholds
- **Section 10.2** (Pricing) — Paid unlock blocking
- **Section 12.5** (Observability) — PSM monitoring
- **Section 12.9** (Internal LLM Architecture) — Kubernetes self-healing, AI tier failover
- **Section 15.0-15.10** (Stress Testing) — PSM validation scenarios

Features

- **F-6.5.7** (Incident Query Engine) — Queryable PSM history
- **F-6.5.13** (Deep Scan) — Blocked in PSM
- **F-6.5.16** (FOCL) — PSM status panel, resource allocation controls
- **F-6.5.17** (Trip Draft Engine) — Allowed in PSM
- **F-6.6.5** (Guardian & Emergency Contact) — Always allowed
- **F-6.6.6** (Emergency Delegate System) — T+48hr notification

External Services

- PagerDuty/Opsgenie — Operator alerting (P1/P2/P3)
- Kubernetes — Infrastructure self-healing
- Prometheus + Grafana — Metrics monitoring

Failure Modes & Safe Defaults

1. PSM Entry Logic Fails

Failure: Trigger detected but PSM activation RPC fails.

Safe Default:

- Fallback: Block ALL mutations (ultra-safe mode)
- Allow read-only operations + evidence export only
- Log failure to backup channel (S3 bucket if ledger unavailable)
- Emit: *psm_activation_failure*

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2. PSM Exit Logic Flaps

Failure: Trigger resolves then immediately re-occurs (flapping).

Safe Default:

- Enforce minimum PSM duration (10 minutes)
- After 3 flaps within 1 hour → Manual approval required to exit
- Emit: *psm_flapping_detected*

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3. Resource Allocation Expires Mid-Operation

Failure: Founder allocates +50% API capacity for 2 hours; allocation expires mid-Deep-Scan.

Safe Default:

- Allow in-flight operations to complete
- Block NEW operations of same type
- Emit: *resource_allocation_expired_mid_operation*

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4. All Regions Enter PSM Simultaneously

Failure: US-East, EU-West, Asia-Pacific all trigger PSM at same time.

Safe Default:

- Auto-escalate to Emergency Mode (Section 8.4.8)
- Founder notification: P1 alert (push + SMS + email simultaneously)
- Standing Containment Extension at T+30min if unreachable
- Emit: *emergency_mode_multi_region_psm*

Security & Privacy Requirements

1. Resource Allocation Logging

All resource allocations logged with:

- Operator ID (who allocated)
- Reason (free-text justification)
- Duration (time-boxed limit)
- Region (which region received resources)
- Timestamp

Immutable: Resource allocation logs cannot be deleted (audit trail).

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2. User Data Protection

PSM does NOT:

- Delete user data
- Modify existing trip data
- Revoke access to evidence
- Block evidence export
- Expire sessions

PSM does:

- Prevent new data mutations that risk corruption
- Preserve all existing data (read-only + export still work)

UX Surfaces

Primary Surfaces (Section 7.3)

1. **PSM Banner** (NEW — top of all screens)
    - Purpose: Inform user of degraded state
    - Actions: Tap to expand details, dismiss
2. **Blocked Operation Modal** (NEW — shown on attempt)
    - Purpose: Explain why action unavailable
    - Actions: Alternative path CTA, "Got it" dismiss
3. **FOCL: Regional Safe Mode Panel** (EXTENDED — F-6.5.16)
    - Purpose: Operator monitoring and resource management
    - Actions: Allocate resources, enable emergency queues, view trigger history
4. **FOCL: Incident Timeline** (NEW — F-6.5.7)
    - Purpose: Queryable audit log of all PSM events
    - Actions: Search by date, filter by region, view auto-recovery actions

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Telemetry

Key Metrics

Metric	Type	Purpose
psm_activations_total	Counter (labeled by region)	How often PSM activated per region
psm_duration_seconds	Histogram (labeled by region)	PSM episode duration distribution
psm_trigger_reasons	Counter (labeled by trigger type, region)	Which triggers most common per region
psm_blocked_operations_total	Counter (labeled by operation type, region)	Which operations users attempted during PSM
psm_resource_allocations_total	Counter (labeled by resource type, region)	How often Founder allocated resources

`psm_flaps_total`	Counter (labeled by region)	How often PSM flapped per region
`psm_auto_recovery_success_rate`	Gauge (labeled by trigger type)	% of PSM episodes that auto-recovered
`psm_emergency_mode_escalations_total`	Counter	How often PSM escalated to Emergency Mode

Test Scenarios

Happy Path

Scenario 1: Regional Surge → PSM Entry → Auto-Recovery

1. US-East traffic spikes to 12,000 incidents/hour (5x normal)
2. Platform enters PSM for US-East region automatically
3. User in US-East sees PSM banner; Deep Scan blocked
4. User in EU-West unaffected (Deep Scan works normally)
5. User in US-East can still create trip, upload evidence, export documentation
6. Traffic subsides to 7,800/hour over 2 hours
7. Recovery validation passes
8. PSM exits after 5-minute cooldown
9. User in US-East sees "Platform fully restored" toast

Expected: Regional degradation isolated; other regions unaffected; auto-recovery successful.

Scenario 2: GPU Server Crash → AI Tier Failover → PSM Avoided

1. GPU Server #2 (US-East-1 cluster) crashes
2. Kubernetes auto-restarts pod (30 seconds)
3. AI Tier 1 temporarily unavailable
4. AI tier failover logic activates → Tier 2 (Gemini Flash)
5. Users experience seamless AI responses (no PSM entry)
6. GPU Server #2 health check passes
7. AI Tier 1 restored
8. Failover reverses → Tier 1 active again

Expected: Infrastructure self-heals; users unaffected; PSM NOT triggered (handled by failover).

Edge Cases

Scenario 3: Resource Allocation Expires Mid-Deep-Scan

1. US-East in PSM (regional surge)
2. Founder allocates +50% API capacity for 2 hours
3. Founder enables emergency Deep Scan queue (100 requests max)
4. User starts Deep Scan (allowed during resource allocation)
5. 1 hour 45 minutes later: Deep Scan still processing
6. Resource allocation expires (2-hour limit)
7. Deep Scan completes successfully (in-flight operation protected)
8. New Deep Scan attempts blocked

Expected: In-flight operation completes; new attempts blocked; no data loss.

Scenario 4: PSM Flapping (Surge Subsides Then Resurges)

1. US-East PSM activates (surge detected: 10,500/hour)
2. Traffic subsides to 7,500/hour
3. PSM exits after 5-minute cooldown
4. 5 minutes later: Traffic resurges to 11,200/hour
5. PSM re-activates
6. Flap #2: PSM exits → re-activates within 10 minutes
7. Flap #3: PSM exits → re-activates within 10 minutes
8. Platform detects flapping → Requires manual approval to exit

Expected: Flapping detected; Founder alerted (P2); manual intervention required.

Scenario 5: Multi-Region PSM → Emergency Mode

1. US-East enters PSM (regional surge)
2. 15 minutes later: EU-West enters PSM (connector cascade: Amadeus + Sabre down)
3. Platform detects ≥2 regions in PSM → Auto-escalate to Emergency Mode
4. Founder notification sent: Push + SMS + Email (simultaneously)
5. Founder unreachable for 30 minutes
6. Standing Containment Extension activates:
    - PSM maintained in both regions
    - New exports queued
    - New registrations held
    - Legal counsel auto-notified

Expected: Emergency escalation correct; Founder reached via all channels; Standing Containment protects platform.

Scenario 6: 48-Hour Founder Unavailable + Conditions Don't Clear

1. US-East enters PSM (ledger write failure)
2. Ledger auto-retry fails (underlying database issue)
3. Emergency Mode triggered (P1 alert)
4. Founder unreachable for 48 hours (traveling, phone off)
5. Standing Containment Extension active at T+30min
6. Legal counsel notified at T+30min
7. Emergency delegate notified at T+48hr (F-6.6.6)
8. Ledger write failure persists (database requires manual DBA intervention)
9. Platform remains in PSM for 48+ hours
10. Core workflows operational (trip creation, evidence upload/export)
11. Founder logs in at T+52hr
12. Reviews FOCL: sees 52-hour PSM episode, trigger reason, auto-recovery attempts
13. Founder contacts DBA, fixes database issue
14. Ledger writes resume
15. Recovery validation passes
16. PSM exits

Expected: Platform survives 52 hours without Founder; core workflows operational; emergency delegate informed; safe default maintained.

Copy Library

Microcopy (Banners, Modals)

Context	Copy
PSM banner (collapsed)	" Safe mode active — Some features temporarily unavailable"
PSM banner (expanded)	"The platform is running in safe mode to protect your data during high load. Core features (trip planning, incident capture, evidence upload, evidence export) are fully functional. Advanced features will return when the platform stabilizes."

Deep Scan blocked	"Deep Scan is temporarily unavailable while we stabilize the platform. Your trip data is safe, and you can still add details manually."
Paid unlock blocked	"Upgrades are paused for now. We've temporarily paused paid upgrades to ensure reliable service. You can still create and edit trips for free."
Claim packet blocked	"Claim packet generation is temporarily unavailable. You can still upload evidence and export your documentation."
PSM exited (toast)	"Platform fully restored — All features available"

Cross-References

Sections

- Section 3.0 (Governance Substrate) — PSM trigger matrix, mutation guards
- Section 3.5 (Trip Maturity States) — PSM mutation guards
- Section 8.4 (Permissioning) — Resource allocation logging
- Section 8.7 (Export Governance) — Export allowed in PSM
- Section 8.8.10 (Founder Reach Protocol) — Emergency notification sequence (T+0: push/SMS/email, T+30min: Standing Containment)
- Section 9.2 (AI Confidence) — AI safety thresholds
- Section 10.2 (Pricing) — Paid unlock blocking
- Section 12.5 (Observability) — PSM monitoring, connector downgrade thresholds
- Section 12.9 (Internal LLM) — Kubernetes self-healing, AI tier failover
- Section 15.0-15.10 (Stress Testing) — PSM validation scenarios

Features

- F-6.5.7 (Incident Query Engine) — Queryable PSM history ("What happened last Thursday?")
- F-6.5.13 (Deep Scan) — Blocked in PSM
- F-6.5.16 (FOCL) — PSM status panel, resource allocation controls
- F-6.5.17 (Trip Draft Engine) — Allowed in PSM
- F-6.5.18 (Route Normalization) — Degraded mode in PSM
- F-6.6.5 (Guardian & Emergency Contact) — Always allowed
- F-6.6.6 (Emergency Delegate System) — T+48hr notification if Founder unreachable

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Feature ID: F-6.6.2 | Protective Safety Mode (PSM) | Version 1.1 (REVISED) | March 2026

Phase 2 Feature | Infrastructure foundation in MVP (Sections 3.0, 8.4, 9.2, 12.5, 12.9) | Full behavioral spec required

Regional scoping: PSM per region | Auto-recovery: When conditions clear + validation + 5min cooldown

48-hour survivability: Platform functions without Founder | Emergency delegation: T+48hr if unreachable

Triggers: Event logging unavailable, regional overload, connector cascade, database sync failure, AI quality degradation, unauthorized changes, cost explosion

Blocked: Deep Scan, paid unlock, claim packets, identity mutations, evidence deletion

Always Allowed: Trip creation, incident capture, evidence upload, ***evidence export****, emergency contact access*

Founder-Friendly FOCL: Plain language narratives, regional status, resource allocation (not "override"), incident timeline